Calculus II: For Physical Sciences and Engineering (AS.110.109)

Lecture: 34

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Textbook Sections: Braun Ch. 2.1, 2.2, 2.3

Todays Topic: An Intro to 2nd order ODEs

Over the last few lectures, we have studied various 1st order ODEs where it is possible to write down explicit formulas for solutions. We also introduced, in homework, the Picard Iteration Scheme which gives a route to proving the existence of a solution to a general 1st order ODE $y' = f(x,y)$, even when there isn't a formula for it.

Today, we will start thinking about 2nd order ODEs. In general, 2nd order ODEs are more difficult to solve than 1st order ODEs, and accordingly posses a more diverse theory. Aside from some generalities today, for the subsequent lectures on the topic we will focus on a special, yet very relevant, subclass of 2nd order ODEs which we can fully analyze.

Def: A 2nd order ODE is an equation of the form

$$y''(t) = F(t, y(t), y'(t))$$

Examples:

1. $y'' = \sin t + y'$
2. $y'' = yy'$
3. $m \cdot y'' = F(t, y, y')$ (Newton's 2nd Law)

(y'': acceleration, F: force, y: position, y': velocity)

Remark: Even just from Example (3) above, we see that 2nd order ODEs have great physical relevance. Other examples include models of oscillation and the trajectories of particles on curved surfaces.

In addition, models involving 2nd order ODEs often come with assumed **initial data** on the solution, for instance in the form

$$\begin{cases} y(t_0) = y_0 \\ y'(t_0) = y_0' \end{cases}$$ ← fixed, given $\mathbb{R}$ numbers

Altogether, an ODE and corresponding initial conditions is called an **initial value problem**.

Example: Recall that, awhile ago, we thought about the calculation of **work** done on a particle by a force, and invoked Newton's 2nd Law

$$F = ma$$

As above, this is actually a 2nd order ODE if we let $a(t) = y''(t)$ be the second derivative of the particles position $y(t)$ at time t. If $y(t)$ = height in meters of a ball of mass 2 kg above the surface of the earth at time t seconds, the following IVP models the height of the ball if we drop it from rest at height 10 m at time $t=0$:

$$\begin{cases} 2 \cdot y''(t) = F_{\text{gravity}}(t, y(t)) = 2 \cdot (-9.8 \text{ m/s}^2) \\ y(0) = 10 \,, \quad y'(0) = 0 \end{cases}$$

If we instead threw the ball upwards at time $t=0$ from height 5 m at a velocity of 2 m/s, then the model would become

$$\begin{cases} 2 \cdot y''(t) = F_{\text{gravity}}(t, y(t)) = 2 \cdot (-9.8 \text{ m/s}^2) \\ y(0) = 5 \,, \quad y'(0) = 2 \end{cases}$$

In any case, this is a basic example of the sort of 2nd order ODEs that we will be able to address in the next few lectures. In particular, we will discuss 2nd order **linear homogeneous** ODEs: of the form

$$y''(t) + p(t)\, y'(t) + q(t)\, y(t) = 0 \qquad (*)$$

where $p(t)$, $q(t)$ are given continuous functions. We'll explain these adjectives in a moment.

As a first result, we have the following (essentially proven using the Picard iteration scheme of last homework applied to systems of ODEs, but this is a topic for another time):

Theorem (Existence and Uniqueness) Suppose p, q are continuous on $t \in (\alpha, \beta)$. Then there exists a unique C^2 function $y(t)$ on (α, β) solving the IVP

$$(*) \quad \begin{cases} y'' + py' + qy = 0 \\ y(0) = y_0, \; y'(0) = y_0' \end{cases}$$

Taking this for granted, we can at least rest assured that studying these equations and their solutions is not going to be in vain!

Before we get to solving such ODEs explicitly (at least in the case where p and q are constant), however, let's spend the rest of the lecture discussing the general properties of such equations.

It will be very useful to cast the ODEs we have been talking about into the language of **operators**.

We think of $\frac{d}{dt}$ and $\frac{d^2}{dt^2}$ as operators (i.e. "functions of functions") taking a function $y(t)$ (which is twice differentiable) to the new functions $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$, respectively.

Abstractly, we denote such **differential operators** by notation like $L[-]$, so that the function $y = y(t)$ is mapped by $L[-]$ to the new function

$$L[y] = L[y](t)$$

Example:

1) $L = \frac{d}{dt}$ acts by: $y(t) \longmapsto L[y](t) = y'(t)$

2) $L = \frac{d^2}{dt^2} + \frac{d}{dt}$ acts by: $y(t) \longmapsto L[y](t) = y''(t) + y'(t)$

3) $L = \frac{d^2}{dt^2} + p(t) \cdot \frac{d}{dt} + q(t) \cdot I$ (identity operator) acts by: $y(t) \longmapsto L[y](t) = y''(t) + p(t)y'(t) + q(t)y(t)$

We can make some pretty complicated operators (relying on suggestive notation to capture the operations we have in mind:

4) $L = \frac{d^3}{dt^3} \cdot \frac{d^2}{dt^2} + \sin(t)\cdot I$ (multiplication, composition) acts by: $y(t) \longmapsto L[y](t) = y'''(t)\cdot y''(t) + \sin(t)\, y(t)$

5) $L = \left(\frac{d}{dt} + I\right) \circ \left(\frac{d}{dt} - I\right)$ acts by: $y(t) \longmapsto L[y](t) = \left(\frac{d}{dt}+I\right)(y'(t) - y(t)) = y''(t) - y(t)$
$= \left(\frac{d^2}{dt^2} - I\right)[y]$

6) $L = \frac{d}{dt} + I\cdot I$ acts by: $y(t) \longmapsto L[y](t) = y'(t) + y(t)^2$

Exercise: Make this explicit - plug some explicit choices of $y(t)$ into the above operators.

eg) if $y(t) = \cos t$ in 6), $L[y](t) = -\sin t + \cos^2 t$.

Amongst all these differential operators, we identify an important subclass with useful properties:

Def: A linear operator is an operator L satisfying

1) $L[y_1 + y_2] = L[y_1] + L[y_2]$ for all functions y_1, y_2

2) $L[cy] = cL[y]$ for all functions y, and all $c \in \mathbb{R}$.

If L is not linear, we call it non-linear.

Exercise: Sort the operators above based on linearity.

eg) Operator (1) is linear

Let's revisit ODE (*) using this language: Here we have $L = \frac{d^2}{dt^2} + p(t)\frac{d}{dt} + q(t)\cdot Id$, so that the ODE takes the form

$$L[y](t) = 0 \qquad (*)$$

L is a linear operator (check this!), which is why we call (*) a linear ODE. We call it homogeneous because of the 0 function on the right hand side of the ODE (if instead of 0 we had some non-identically zero function $g(t)$, then (*) would be called inhomogeneous.

Next time, we'll talk about finding solutions to problems like (*). As a sneak peak, what we will find is the following (which is entirely due to the fact that the problem is linear and homogeneous):

1) If $y_1(t)$ solves (*), then so does $cy_1(t)$ for any $c \in \mathbb{R}$

2) If $y_1(t)$ and $y_2(t)$ both solve (*), then so does $y_1(t) + y_2(t)$

   Thus, every $y(t) = \alpha y_1(t) + \beta y_2(t)$, $\alpha, \beta \in \mathbb{R}$, solves (*)

3) It is possible to find two "independent" solutions $y_1(t), y_2(t)$ to (*) so that every possible solution to (*) is of the form $y(t) = \alpha y_1(t) + \beta y_2(t)$ for some $\alpha, \beta \in \mathbb{R}$.

4) Once we have such a "basis set of fundamental solutions," we can uniquely solve every IVP

$$\begin{cases} L[y](t) = y''(t) + p(t)y'(t) + q(t)y(t) = 0 \\ y(0) = y_0 \quad y'(0) = y_0'. \end{cases}$$

by solving for the <u>two</u> unknown coefficients α, β using the <u>two</u> initial conditions.